

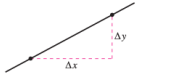
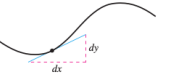
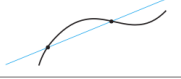
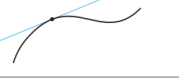
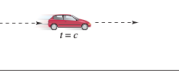
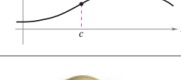
1.1 Introducing Calculus: Can Change Occur at an Instant?

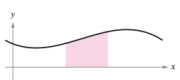
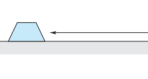
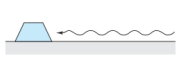
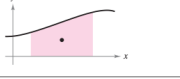
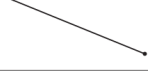
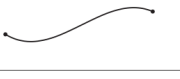
What is Calculus?

Calculus is the mathematics of **change**.

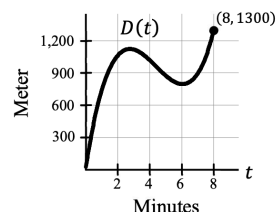
Calculus is the mathematics of velocities, accelerations, tangent lines, slopes, areas, volumes, arc lengths, centroids, curvatures, and a variety of other concepts that have enabled scientists, engineers, and economists to model real-life situations.

Although precalculus mathematics also deals with velocities, accelerations, tangent lines, slopes, and so on, there is a fundamental difference between precalculus mathematics and calculus. Precalculus mathematics is more static, whereas calculus is more dynamic.

Without Calculus	With Differential Calculus
Value of $f(x)$ when $x = c$ 	Limit of $f(x)$ as x approaches c 
Slope of a line 	Slope of a curve 
Secant line to a curve 	Tangent line to a curve 
Average rate of change between $t = a$ and $t = b$ 	Instantaneous rate of change at $t = c$ 
Curvature of a circle 	Curvature of a curve 
Height of a curve when $x = c$ 	Maximum height of a curve on an interval 
Tangent plane to a sphere 	Tangent plane to a surface 
Direction of motion along a line 	Direction of motion along a curve 

Without Calculus	With Integral Calculus
Area of a rectangle 	Area under a curve 
Work done by a constant force 	Work done by a variable force 
Center of a rectangle 	Centroid of a region 
Length of a line segment 	Length of an arc 
Surface area of a cylinder 	Surface area of a solid of revolution 
Mass of a solid of constant density 	Mass of a solid of variable density 
Volume of a rectangular solid 	Volume of a region under a surface 
Sum of a finite number of terms $a_1 + a_2 + \dots + a_n = S$	Sum of an infinite number of terms $a_1 + a_2 + a_3 + \dots = S$

Ex 1: Mr. Packer's distance from his house is modeled by the function $D(t)$. While riding his bike to the store, he realizes he dropped his wallet and turns around to find it. After finding his wallet, he finishes his ride to the store.



a) Calculate Mr. Packer's average speed (rate of change) for his trip to the store if he arrives after 8 minutes. Indicated units of measure.

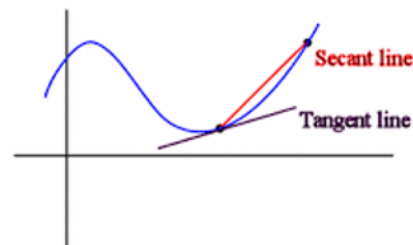
$$\frac{D(8)-D(0)}{8-0} = \frac{1300}{8} \text{ meters/min}$$

b) Was Mr. Packer's average rate of change greater between 2 and 6 minutes or between 2 and 3 minutes? Give a reason for your answer.

$\frac{D(6)-D(2)}{6-2} < 0$ and $\frac{D(3)-D(2)}{3-2} > 0$, therefore Mr. Packer's average rate of change is greater between 2 and 3 minutes.

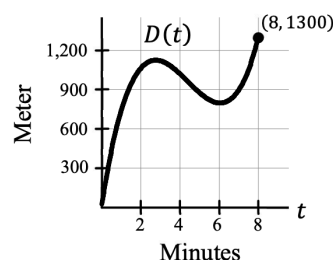
Average Rate vs. Instantaneous Rate

Average rate of change is calculated on an interval, while instantaneous rate of change is calculated at a single point. Graphically, average rate of change represents the slope of a secant line and instantaneous rate of change represents the slope of a tangent line.



The instantaneous rate of change can be estimated by computing the average rate of change over a short interval.

Ex 1: Mr. Packer's distance from his house is modeled by the function $D(t)$. While riding his bike to the store, he realizes he dropped his wallet and turns around to find it. After finding his wallet, he finishes his ride to the store.



c) Write, but do not evaluate, an expression to give a close estimate of the instantaneous rate of change at $t = 2$.

$$\frac{D(2) - D(1.999)}{2 - 1.999}$$

**answers will vary*

d) Interpret the meaning of $\frac{D(4) - D(3.999)}{4 - 3.999}$ in the context of the problem.

$\frac{D(4) - D(3.999)}{4 - 3.999}$ is an estimate of the speed in meters/minute at which Mr. Packer rides his bicycle at $t = 4$ minutes.

**Adapted from The Algebras*

Ex 2: 2025 AB 3A

A student starts reading a book at time $t = 0$ minutes and continues reading for the next 10 minutes. The rate at which the student reads is modeled by the differentiable function R , where $R(t)$ is measured in words per minute. Selected values of $R(t)$ are given in the table shown.

t (minutes)	0	2	8	10
$R(t)$ (words per minute)	90	100	150	162

$$R'(1) \approx \frac{R(2) - R(0)}{2 - 0}$$

$$R'(1) \approx \frac{100 - 90}{2 - 0}$$

$R'(1) \approx 5$ words per minute per minute

A. Approximate $R'(1)$ using the average rate of change of R over the interval $0 \leq t \leq 2$. Show the work that leads to your answer. Indicate units of measure.

Ex 3:

Let $f(x) = (x + 2)^2$. Put the following in order from least to greatest.

- The average rate of change over the interval $[-5, 1]$
- The average rate of change over the interval $[-3, 4]$
- The instantaneous rate of change at $x = -3$
- The instantaneous rate of change at $x = 4$

$$a < c < b < d$$

**Math Medic*

1.2 - 1.4

AP Calculus AB Unit 1 Notes

1.2 Defining Limits and Using Limit Notation

1.3 Estimating Limit Values from Graphs

1.4 Estimating Limit Values from Tables

What is a Limit?

In mathematics, a limit is the y-value a curve approaches as its input approaches a certain value. *"Limits are about the journey, not the destination!"*

Limits are fundamental in calculus and mathematical analysis, used to define concepts like derivatives, integrals, and continuity.

Limits can be found graphically, numerically, and algebraically (analytically).



Limit Notation

When you take your AP Calculus exam in May you will be graded based on how well you communicate your answers.



$$\lim_{x \rightarrow 5} f(x) = 17$$

$$\lim_{x \rightarrow -\infty} e^x = 0$$

Communication and Notation 4

Use correct notation, language, and mathematical conventions to communicate results or solutions.

4.A Use precise mathematical language.

4.B Use appropriate units of measure.

4.C Use appropriate mathematical symbols and notation (e.g., Represent a derivative using $f'(x)$, y' , and $\frac{dy}{dx}$).

4.D Use appropriate graphing techniques.

4.E Apply appropriate rounding procedures.



$$\lim_{x \rightarrow 7} = 8$$

As you begin your calculus journey, make sure to consistently practice using good notation!

Math Speak

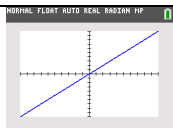
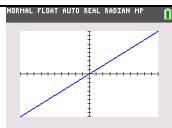
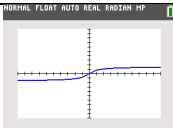
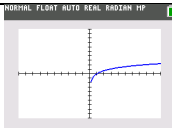
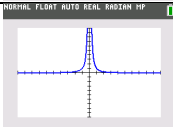
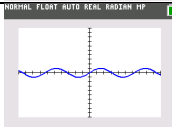
Limits either exist or do not exist.

- If a limit exists, write the numerical value of the limit: $\lim_{x \rightarrow 5} h(x) = 10$
- If a limit does not exist, use the abbreviation DNE: $\lim_{t \rightarrow \infty} g(t)$ DNE

In words, $\lim_{x \rightarrow 5} h(x) = 10$ is read "the limit of $h(x)$ as x approaches 5 is equal to 10."

Finding Limits Graphically

Ex 1: Find the limit graphically. If the limit does not exist, explain.

a) $\lim_{x \rightarrow \infty} x = \infty$		b) $\lim_{x \rightarrow -\infty} x = -\infty$	
c) $\lim_{x \rightarrow -\infty} \tan^{-1} x = -\frac{\pi}{2}$		d) $\lim_{x \rightarrow \infty} \ln x = \infty$	
e) $\lim_{x \rightarrow \infty} \frac{1}{x^2} = 0$		f) $\lim_{x \rightarrow \infty} \sin x$ DNE, $\sin x$ oscillates between -1 and 1 as $x \rightarrow \infty$.	

1.2 - 1.4

One-Sided Limits

When finding a limit as x approaches a finite value c it is essential to determine how the y -values on each side of c behave.

Left-Hand Limit

$$\lim_{x \rightarrow c^-} f(x)$$

Right-Hand Limit

$$\lim_{x \rightarrow c^+} f(x)$$

The $\lim_{x \rightarrow c} f(x)$ exists if and only if

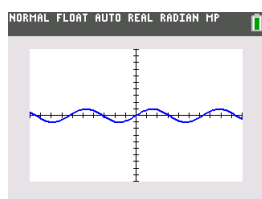
$$\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x).$$



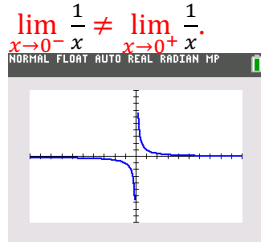
When finding the limit "from the left" does not mean you are moving left! Simply, start from the left (think "less than") and head towards the x -value the limit is approaching.

Ex 2: Find the limit graphically. If the limit does not exist, explain.

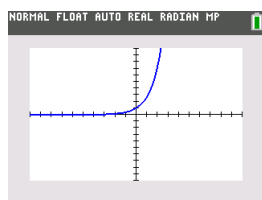
g) $\lim_{x \rightarrow 0} \sin x = 0$



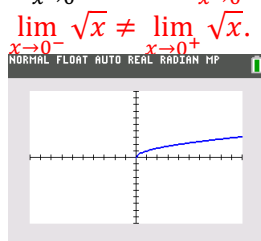
h) $\lim_{x \rightarrow 0} \frac{1}{x}$ DNE, $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$ and $\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$



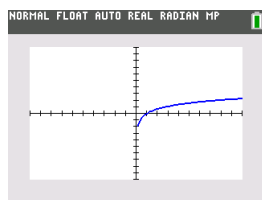
i) $\lim_{x \rightarrow 0} e^x = 1$



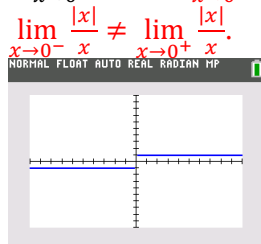
j) $\lim_{x \rightarrow 0} \sqrt{x}$ DNE, $\lim_{x \rightarrow 0^-} \sqrt{x}$ DNE and $\lim_{x \rightarrow 0^+} \sqrt{x} = 0$



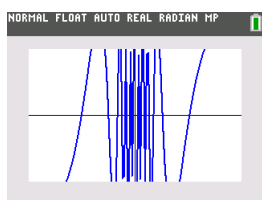
k) $\lim_{x \rightarrow 1} \ln x = 0$



l) $\lim_{x \rightarrow 0} \frac{|x|}{x}$ DNE, $\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$ and $\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$



Ex 3: Evaluate $\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$. If the limit does not exist, explain.

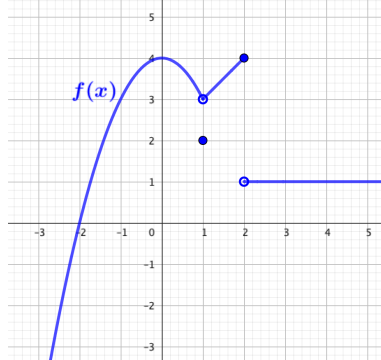


$\lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right)$ DNE. The graph of $\sin\left(\frac{1}{x}\right)$ oscillates at $x = 0$.

1.2 - 1.4

AP Calculus AB Unit 1 Notes

Ex 4: Using the graph of $f(x)$ find each limit graphically. If the limit does not exist, explain.



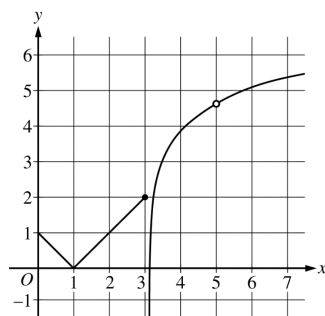
a) $\lim_{x \rightarrow \infty} f(x) = 1$

b) $\lim_{x \rightarrow -\infty} f(x) = -\infty$

c) $\lim_{x \rightarrow 0} f(x) = 4$

d) $\lim_{x \rightarrow 1} f(x) = 3$

Ex 5:



Graph of f

The graph of a function f is shown above. Which of the following limits does not exist?

(A) $\lim_{x \rightarrow 1^-} f(x)$

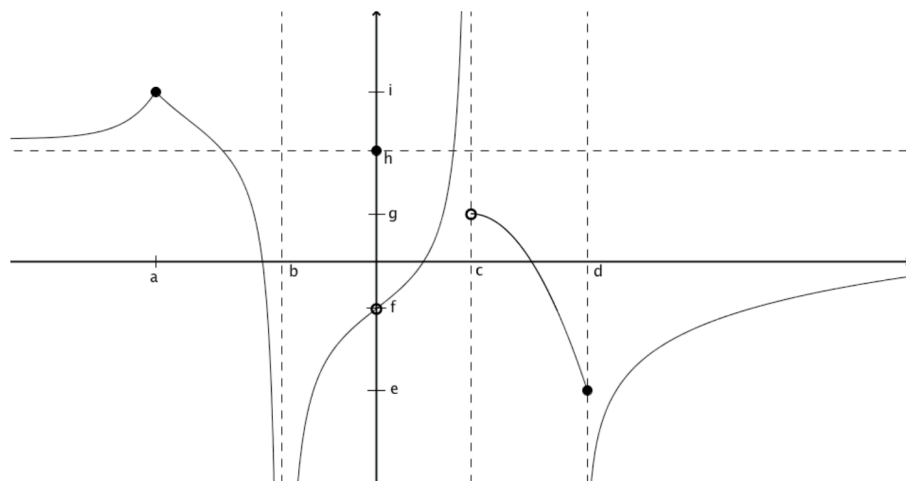
(B) $\lim_{x \rightarrow 1} f(x)$

(C) $\lim_{x \rightarrow 3} f(x)$

(D) $\lim_{x \rightarrow 5} f(x)$

(E) $\lim_{x \rightarrow 5} f(x)$

Ex 6: The graph below represents the function $g(x)$. The function $g(x)$ has vertical asymptotes at $x = b, x = c$ and $x = d$. The function $g(x)$ has a horizontal asymptote at $y = h$. Evaluate each expression. If a limit does not exist, explain.



a) $\lim_{x \rightarrow a^-} g(x) = i$

b) $\lim_{x \rightarrow a^+} g(x) = i$

c) $\lim_{x \rightarrow a} g(x) = i$

d) $\lim_{x \rightarrow \infty} g(x) = h$

e) $\lim_{x \rightarrow -\infty} g(x) = h$

f) $\lim_{x \rightarrow b^-} g(x) = -\infty$

g) $\lim_{x \rightarrow b^+} g(x) = -\infty$

h) $\lim_{x \rightarrow b} g(x) = -\infty$

i) $\lim_{x \rightarrow c} g(x)$ DNE,
 $\lim_{x \rightarrow c^-} g(x) = \infty \neq$
 $\lim_{x \rightarrow c^+} g(x) = g$

j) $\lim_{x \rightarrow 0} g(x) = f$

k) $\lim_{x \rightarrow d} g(x)$ DNE,
 $\lim_{x \rightarrow d^-} g(x) = e \neq$
 $\lim_{x \rightarrow d^+} g(x) = -\infty$

l) $g(0) = h$

m) $g(a) = i$

n) $g(d) = e$

o) $g(c)$ undefined

p) $g(b)$ undefined

AP Calculus AB Unit 1 Notes

Limits can be found numerically by constructing a table of values. Using a graphing utility, pick at least three x -values that get progressively “closer” to $x = c$.



*always round (or truncate) to three places after the decimal unless otherwise indicated.



Plot1 Plot2 Plot3
 $Y_1 = \frac{x^4}{\sqrt{x^2-1}}$
 $Y_2 =$
 $Y_3 =$
 $Y_4 =$
 $Y_5 =$
 $Y_6 =$
 $Y_7 =$

X	Y1
100	5.77360
1000	5.77108
10000	5.8E11
100000	5.8E14

Y=

[illegible]

NORMAL FLOAT AUTO REAL RADIAN HP				NORMAL FLOAT AUTO REAL RADIAN HP			
Plot1	Plot2	Plot3		X	Y1		
	$Y_1 = X+3$			-100	97		
	$Y_2 = 14X-2$			-1000	1402		
				-10000	0.0712		
				-100000	0.0714		
				-1E5	0.0714		
							
							
							
							

NORMAL FLOAT AUTO REAL RADIAN HP			NORMAL FLOAT AUTO REAL RADIAN HP		
Plot1	Plot2	Plot3	X	Y1	
	$\frac{5-X}{3X^2-4}$		-100	0.0035	
			-1000	3.4E-4	
			-10000	3.3E-5	
			-1E5	3.3E-6	

X=

Figure 1 shows two screenshots of the TI-84 Plus calculator interface. The left screenshot displays the Plot1 window with the equation $Y1 = \frac{X^2 - 3X + 2}{X - 2}$ entered. The right screenshot displays the Y= window with the same equation entered as Y1, and the X and Y axes labeled.

Left screenshot: The display shows 'Plot1 Plot2 Plot3' at the top, followed by the function 'ln(10)' and the result '2.302585'.

Right screenshot: The display shows the result '2.302585'.

NORMAL FLOAT AUTO REAL RADIAN MP NORMAL FLOAT AUTO REAL RADIAN MP NORMAL FLOAT AUTO REAL RADIAN MP

MATH NUM CMPLX PROB FRAC Plot1 Plot2 Plot3 X Y1

1: abs($\text{Y1} = \text{int}(X)$ 1.9
 2: round($\text{Y2} =$ 1.99
 3: iPart($\text{Y3} =$ 1.999
 4: fPart($\text{Y4} =$
 5: int($\text{Y5} =$
 6: min($\text{Y6} =$
 7: max($\text{Y7} =$
 8: lcm($\text{Y8} =$
 9: gcd($\text{Y9} =$
 X=

1.2 - 1.4

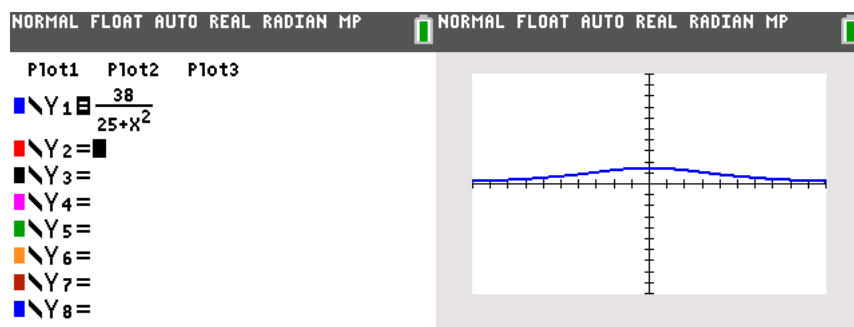
AP Calculus AB Unit 1 Notes

Ex 8: 2025 AB 1C

An invasive species of plant appears in a fruit grove at time $t = 0$ and begins to spread. The function C defined by $C(t) = 7.6 \arctan(0.2t)$ models the number of acres in the fruit grove affected by the species t weeks after the species appears. It can be shown that $C'(t) = \frac{38}{25 + t^2}$.

C. Assume that the invasive species continues to spread according to the given model for all times $t > 0$. Write a limit expression that describes the end behavior of the rate of change in the number of acres affected by the species. Evaluate this limit expression.

$$\lim_{t \rightarrow \infty} C'(t) = 0$$



When Limits DNE...

In general, limits do not exist when

1. $f(x)$ approaches **two different** y -values from the left and the right of $x = c$.
2. $f(x)$ **oscillates** between two or more y -values at $x = c$.
3. $f(x)$ approaches ∞ or $-\infty$ * (We will explore this further later in Unit 1)

Explanations and What To Write:

If c is finite:

$\lim_{x \rightarrow c} f(x)$ DNE because

$$\lim_{x \rightarrow c^-} f(x) \neq \lim_{x \rightarrow c^+} f(x)$$

If c is infinite:

$\lim_{x \rightarrow c} f(x)$ DNE because

$f(x)$ approaches ∞ or $-\infty$

OR

$f(x)$ oscillates at $x = c$

1.5 - 1.7

1.5 Determining Limits Using Algebraic Properties of Limits

1.6 Determining Limits Using Algebraic Manipulation

1.7 Selecting Procedures for Determining Limits

Algebraic Properties of Limits

Let b and c be real numbers, let n be a positive integer, and let f and g be functions with the limits $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} g(x) = K$.

1. Scalar multiple: $\lim_{x \rightarrow c} [bf(x)] = \underline{bL}$
2. Sum or difference: $\lim_{x \rightarrow c} [f(x) \pm g(x)] = \underline{L \pm K}$
3. Product: $\lim_{x \rightarrow c} [f(x)g(x)] = \underline{LK}$
4. Quotient: $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \underline{\frac{L}{K}}, K \neq 0$
5. Power: $\lim_{x \rightarrow c} [f(x)]^n = \underline{L^n}$

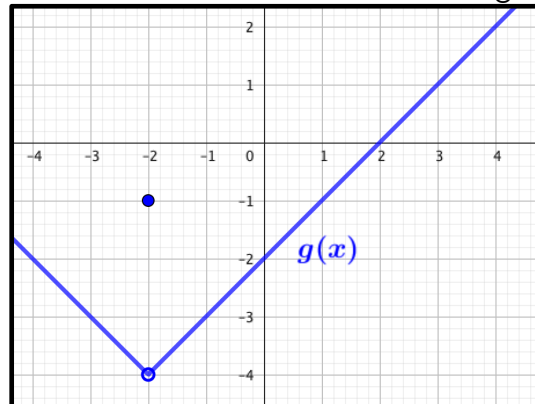
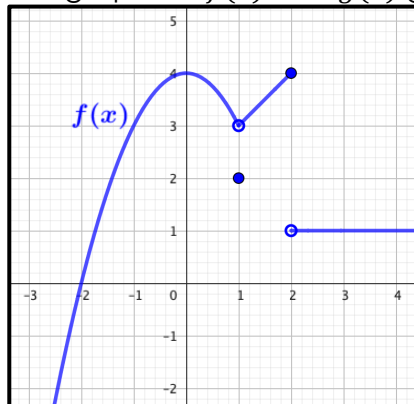


Be careful! If either $\lim_{x \rightarrow c} f(x)$ or $\lim_{x \rightarrow c} g(x)$ DNE, these limit properties CANNOT be used!

Ex 1: If $\lim_{x \rightarrow c} f(x) = 4$ and $\lim_{x \rightarrow c} g(x) = 10$, evaluate the following limits.

a) $\lim_{x \rightarrow c} f(x)g(x)$ $4 \cdot 10 = 40$ *answers need not be simplified	b) $\lim_{x \rightarrow c} 7[f(x)]^{3/2}$ $7[4]^{3/2} = 56$	c) $\lim_{x \rightarrow c} [g(x) - f(x)]$ $10 - 4 = 6$
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Ex 2: Consider the graphs of $f(x)$ and $g(x)$ given below. Find each of the following limits, if possible.



a) $\lim_{x \rightarrow 0} [f(x) + g(x)]$ $4 - 2 = 2$	b) $\lim_{x \rightarrow -2} \frac{4f(x)}{g(x)}$ $\frac{4(0)}{-4} = 0$
c) $\lim_{x \rightarrow -1} [f(x)]^2$ $[3]^2 = 9$	d) $\lim_{x \rightarrow 2} [f(x)g(x)]$ $\lim_{x \rightarrow 2^-} [f(x)g(x)] = (4)(0) = 0$ $\lim_{x \rightarrow 2^+} [f(x)g(x)] = (2)(0) = 0$ $\lim_{x \rightarrow 2} [f(x)g(x)] = 0$

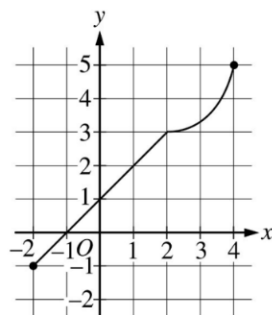
1.5 - 1.7

AP Calculus AB Unit 1 Notes

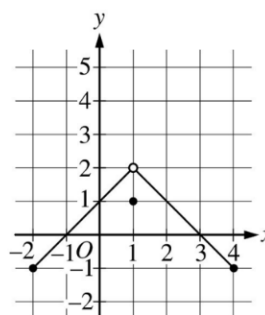
Algebraic Properties of Composite Limits

If $\lim_{x \rightarrow c} f(g(x)) = f\left(\lim_{x \rightarrow c} g(x)\right)$, provided $\lim_{x \rightarrow c} g(x) = L$ and $f(x)$ is continuous at $x = L$.

Ex 3:



Graph of f



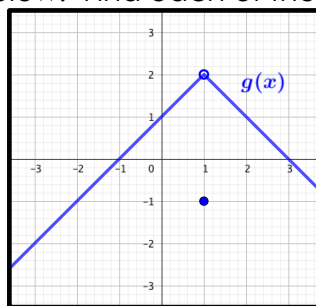
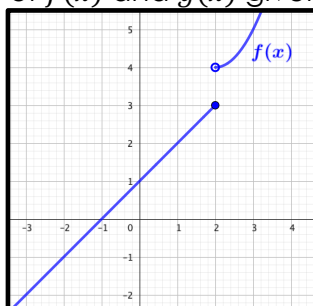
Graph of g

$$\begin{aligned} &f\left(\lim_{x \rightarrow 1} g(x)\right) \\ &f(2) \\ &3 \end{aligned}$$

The graphs of the functions f and g are shown above. The value of $\lim_{x \rightarrow 1} f(g(x))$ is

- (A) 1
- (B) 2
- (C) 3**
- (D) nonexistent

Ex 4: Consider the graphs of $f(x)$ and $g(x)$ given below. Find each of the following limits, if possible.



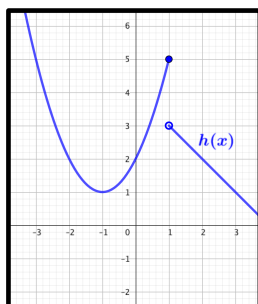
a) $\lim_{x \rightarrow 2} g(f(x))$

$$\begin{aligned} \lim_{x \rightarrow 2^-} f(x) &= 3 \\ \lim_{x \rightarrow 2^+} f(x) &= 4 \\ \lim_{x \rightarrow 2} g(f(x)) &\text{ DNE} \end{aligned}$$

b) $\lim_{x \rightarrow 1} f(g(x))$

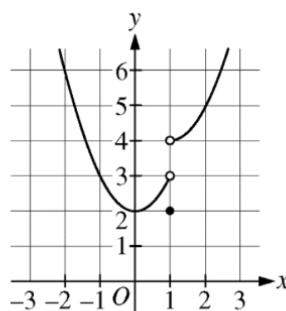
$$\begin{aligned} \lim_{x \rightarrow 1} g(x) &= 2^- \\ \lim_{x \rightarrow 2^-} f(x) &= 3 \\ \lim_{x \rightarrow 1} f(g(x)) &= 3 \end{aligned}$$

Ex 5: Consider the graph of $h(x)$. Evaluate $\lim_{x \rightarrow -1} h(h(x))$



$$\begin{aligned} \lim_{x \rightarrow -1} h(x) &= 1^+ \\ \lim_{x \rightarrow 1^+} h(x) &= 3 \\ \lim_{x \rightarrow 1} f(g(x)) &= 3 \end{aligned}$$

Ex 6:



$$\begin{aligned}\lim_{x \rightarrow 0} (1 - x^2) &= 1^- \\ \lim_{x \rightarrow 1^-} f(x) &= 3 \\ \lim_{x \rightarrow 0} f(1 - x^2) &= 3\end{aligned}$$

The graph of the function f is shown in the figure above. The value of $\lim_{x \rightarrow 0} f(1 - x^2)$ is

- (A) 1 (B) 2 (C) 3 (D) 4 (E) nonexistent

Evaluating Limits Algebraically – Direct Substitution

Ex 7: Evaluate, if possible.

a) $\lim_{x \rightarrow 2} (5x^2 - 1)$ 19	b) $\lim_{\theta \rightarrow \frac{\pi}{3}} \sin \theta$ $\frac{\sqrt{3}}{2}$	c) $\lim_{t \rightarrow 0} \frac{t^2 - 1}{t - 1}$ 1	d) $\lim_{t \rightarrow 1} \frac{t^2 - 1}{t - 1}$ *can't do....YET!
--	--	--	--

Indeterminate Forms

An indeterminate form is a mathematical expression whose value cannot be determined. There are seven indeterminate forms but only two are tested in AP Calculus: $\frac{0}{0}$ and $\pm \frac{\infty}{\infty}$.



Never set a limit equal to an indeterminate form! If direct substitution leads to an indeterminate form, choose an alternate way to find the limit.

Techniques For Dealing with Indeterminate Forms

Do not stop when direct substitution leads to an indeterminate form. Instead, algebraically manipulate the expression, simplify, then take the limit again. The following techniques can be used to simplify the expression:

1. Factoring
2. Rationalizing
3. Trigonometric Identities
4. Simplifying a Complex Fraction

Ex 8: Find the limit, if possible.

a) $\lim_{t \rightarrow 1} \frac{t^2 - 1}{t - 1}$ $\lim_{t \rightarrow 1} \frac{(t + 1)(\cancel{t - 1})}{\cancel{t - 1}}$ $\lim_{t \rightarrow 1} (t + 1)$ 2	b) $\lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h}$ $\lim_{h \rightarrow 0} \frac{(x + h - x)((x + h)^2 + x(x + h) + x^2)}{h}$ $\lim_{h \rightarrow 0} \frac{\cancel{h}((x + h)^2 + x(x + h) + x^2)}{\cancel{h}}$ $\lim_{h \rightarrow 0} ((x + h)^2 + x(x + h) + x^2)$ $3x^2$
--	---

1.5 - 1.7

AP Calculus AB Unit 1 Notes

c) $\lim_{x \rightarrow 0} \frac{\sqrt{x+1}-1}{x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sqrt{x+1}-1}{x} &= \lim_{x \rightarrow 0} \frac{\sqrt{x+1}-1}{x} \cdot \frac{\sqrt{x+1}+1}{\sqrt{x+1}+1} \\ &= \lim_{x \rightarrow 0} \frac{x+1-1}{x(\sqrt{x+1}+1)} \\ &= \lim_{x \rightarrow 0} \frac{x}{x(\sqrt{x+1}+1)} \\ &= \lim_{x \rightarrow 0} \frac{1}{\sqrt{x+1}+1} \\ &= \frac{1}{2} \end{aligned}$$

d) $\lim_{x \rightarrow 0} \frac{\frac{1}{x+4} - \frac{1}{4}}{x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\frac{1}{x+4} - \frac{1}{4}}{x} &= \lim_{x \rightarrow 0} \frac{\frac{1}{x+4} - \frac{1}{4}}{x} \cdot \frac{4(x+4)}{4(x+4)} \\ &= \lim_{x \rightarrow 0} \frac{4 - (x+4)}{4x(x+4)} \\ &= \lim_{x \rightarrow 0} \frac{-x}{4x(x+4)} \\ &= \lim_{x \rightarrow 0} \frac{-1}{4(x+4)} \\ &= -\frac{1}{16} \end{aligned}$$

e) $\lim_{x \rightarrow \frac{3\pi}{2}} \frac{\sin 2x}{\cos x}$

$$\begin{aligned} \lim_{x \rightarrow \frac{3\pi}{2}} \frac{\sin 2x}{\cos x} &= \lim_{x \rightarrow \frac{3\pi}{2}} \frac{2 \sin x \cos x}{\cos x} \\ &= \lim_{x \rightarrow \frac{3\pi}{2}} 2 \sin x \\ &= 2 \sin \frac{3\pi}{2} \\ &= -2 \end{aligned}$$

f) $\lim_{x \rightarrow \frac{\pi}{8}} (\sin^2 x + \cos^2 x)$

$$\lim_{x \rightarrow \frac{\pi}{8}} 1 = 1$$

g) $\lim_{x \rightarrow 0} \sin x \cot x$

$$\begin{aligned} \lim_{x \rightarrow 0} \sin x \cot x &= \lim_{x \rightarrow 0} \sin x \cdot \frac{\cos x}{\sin x} \\ &= \lim_{x \rightarrow 0} \cos x \\ &= \cos 0 \\ &= 1 \end{aligned}$$

h) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos^2 x}{1 - \sin x}$

$$\begin{aligned} \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos^2 x}{1 - \sin x} &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin^2 x}{1 - \sin x} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{(1 + \sin x)(1 - \sin x)}{1 - \sin x} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 + \sin x}{1} \\ &= 1 + \sin \frac{\pi}{2} \\ &= 2 \end{aligned}$$

Limits of Piecewise Functions

Ex 9: Find the limit, if possible.

a) $\lim_{x \rightarrow 0} f(x), \quad f(x) = \begin{cases} \sin x, & x < 0 \\ 1 - \cos x, & 0 \leq x \leq \pi \\ \cos x, & x > \pi \end{cases}$

$$\begin{aligned} \lim_{x \rightarrow 0^-} \sin x &= 0 \\ \lim_{x \rightarrow 0^+} (1 - \cos x) &= 0 \\ \lim_{x \rightarrow 0} f(x) &= 0 \end{aligned}$$

b) $\lim_{x \rightarrow \pi} f(x), \quad f(x) = \begin{cases} \sin x, & x < 0 \\ 1 - \cos x, & 0 \leq x \leq \pi \\ \cos x, & x > \pi \end{cases}$

$$\begin{aligned} \lim_{x \rightarrow \pi^-} (1 - \cos x) &= 2 \\ \lim_{x \rightarrow \pi^+} \cos x &= -1 \\ \lim_{x \rightarrow \pi} f(x) &\text{DNE} \end{aligned}$$

c) $\lim_{x \rightarrow \frac{4\pi}{3}} f(x), \quad f(x) = \begin{cases} \sin x, & x < 0 \\ 1 - \cos x, & 0 \leq x \leq \pi \\ \cos x, & x > \pi \end{cases}$

$$\begin{aligned} \lim_{x \rightarrow \frac{4\pi}{3}} \cos x &= \cos \frac{4\pi}{3} \\ &= -\frac{1}{2} \end{aligned}$$

d) $\lim_{x \rightarrow -5} g(x), \quad g(x) = \begin{cases} \frac{x^2-25}{x+5}, & x \neq -5 \\ 7, & x = -5 \end{cases}$

$$\begin{aligned} \lim_{x \rightarrow -5} \frac{x^2-25}{x+5} &= \lim_{x \rightarrow -5} \frac{(x-5)(x+5)}{x+5} \\ &= \lim_{x \rightarrow -5} (x-5) \\ &= -10 \end{aligned}$$

1.8 - 1.9

AP Calculus AB Unit 1 Notes

1.8 Determining Limits Using the Squeeze Theorem

1.9 Connecting Multiple Representations of Limits

Special Trigonometric Limits

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$$

*these limits can be proved using "The Squeeze Theorem" or L'Hopital's Rule.

Ex 1: Find the limit, if possible.

a) $\lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{1 - \cos 4x}{x} &\cdot \frac{4}{4} \\ \lim_{x \rightarrow 0} \frac{1 - \cos 4x}{4x} &\cdot \frac{1}{4} \\ 0 &\cdot \frac{1}{4} \\ 0 \end{aligned}$$

b) $\lim_{x \rightarrow 0} \frac{\sin x}{6x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin x}{x} &\cdot \frac{1}{6} \\ 1 &\cdot \frac{1}{6} \\ \frac{1}{6} \end{aligned}$$

c) $\lim_{x \rightarrow 0} \frac{x}{\sin x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{-1} \\ 1^{-1} \\ 1 \end{aligned}$$

d) $\lim_{x \rightarrow 0} \frac{\sin 3x}{5x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin 3x}{5x} &\cdot \frac{3}{3} \\ \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} &\cdot \frac{3}{5} \\ 1 &\cdot \frac{3}{5} \\ \frac{3}{5} \end{aligned}$$

e) $\lim_{x \rightarrow 0} \frac{\tan x}{x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin x}{\cos x} \\ \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{1}{\cos x} \\ 1 \cdot 1 \\ 1 \end{aligned}$$

f) $\lim_{x \rightarrow 0} \frac{(1 - \cos x)^2}{x}$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{(1 - \cos x)(1 - \cos x)}{x} \\ \lim_{x \rightarrow 0} \frac{1 - \cos x}{x} \cdot (1 - \cos x) \\ 0 \cdot 0 \\ 0 \end{aligned}$$

Ex 2:

$$\lim_{x \rightarrow 0} \frac{1 - \cos^2(2x)}{(2x)^2} =$$

(A) 0

(B) $\frac{1}{4}$

(C) $\frac{1}{2}$

(D) 1

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin^2(2x)}{(2x)^2} \\ \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \right)^2 \\ (1)^2 \\ 1 \end{aligned}$$

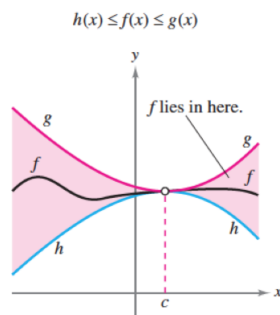
The Squeeze Theorem

THEOREM 1.8 The Squeeze Theorem

If $h(x) \leq f(x) \leq g(x)$ for all x in an open interval containing c , except possibly at c itself, and if

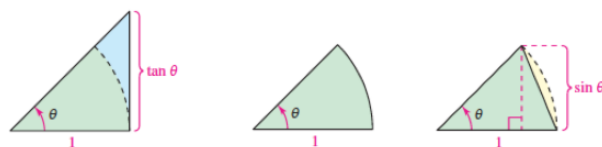
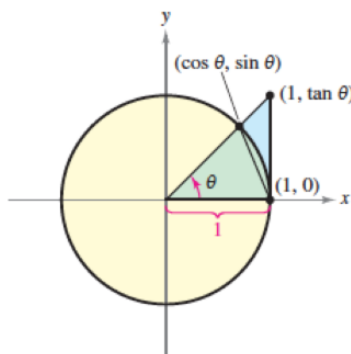
$$\lim_{x \rightarrow c} h(x) = L = \lim_{x \rightarrow c} g(x)$$

then $\lim_{x \rightarrow c} f(x)$ exists and is equal to L .



The Squeeze Theorem – Proof

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$



Area of triangle	$\geq$	Area of sector	$\geq$	Area of triangle
$\frac{1}{2}(1)(\tan \theta)$	$\geq$	$\frac{1}{2}(1^2)(\theta)$	$\geq$	$\frac{1}{2}(1)(\sin \theta)$
$\frac{\tan \theta}{2} \left(\frac{2}{\sin \theta} \right)$	$\geq$	$\frac{\theta}{2} \left(\frac{2}{\sin \theta} \right)$	$\geq$	$\frac{\sin \theta}{2} \left(\frac{2}{\sin \theta} \right)$
$\cos \theta$	$\geq$	$\frac{\theta}{\sin \theta}$	$\geq$	1
$\frac{1}{\cos \theta}$	$\leq$	$\frac{\sin \theta}{\theta}$	$\leq$	1
$\lim_{x \rightarrow 0} \frac{1}{\cos \theta}$	$\leq$	$\lim_{x \rightarrow 0} \frac{\sin \theta}{\theta}$	$\leq$	$\lim_{x \rightarrow 0} 1$

$$1 \leq \lim_{x \rightarrow 0} \frac{\sin \theta}{\theta} \leq 1$$

By the Squeeze Theorem, $\lim_{x \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

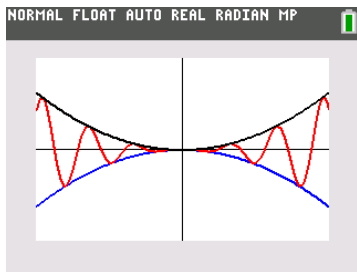
1.8 - 1.9

AP Calculus AB Unit 1 Notes

Ex 3: Let $h(x) = -x^2$, $f(x) = x^2 \cdot \cos(10\pi x)$, and $g(x) = x^2$

a) Use a graphing utility to show

$$h(x) \leq f(x) \leq g(x).$$



b) Use the squeeze theorem to evaluate $\lim_{x \rightarrow 0} f(x)$.

$$\lim_{x \rightarrow 0} h(x) \leq \lim_{x \rightarrow 0} f(x) \leq \lim_{x \rightarrow 0} g(x)$$

$$0 \leq \lim_{x \rightarrow 0} f(x) \leq 0$$

$$\text{By the Squeeze Theorem, } \lim_{x \rightarrow 0} f(x) = 0$$

*Inspired by Tony Record

Four Corners

$$f(x) = x^2 - 3x - 28$$

$$g(x) = x + 4$$

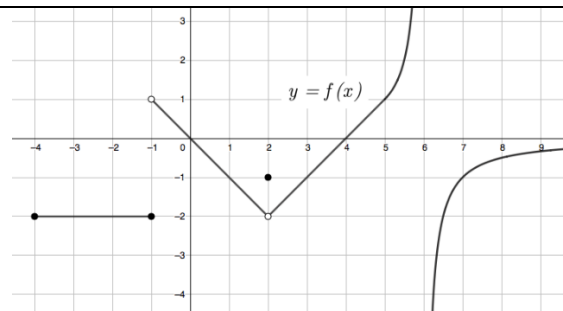
$$h(x) = -2|x + 4|$$

Use the functions above to evaluate the following limits. If a limit does not exist, write DNE.

$$1. \lim_{x \rightarrow -4} \frac{f(x)}{g(x)} = \lim_{x \rightarrow -4} \frac{(x-7)(x+4)}{(x+4)} = -11$$

$$2. \lim_{x \rightarrow 10} \frac{g(x)}{f(x)} = \lim_{x \rightarrow 10} \frac{(x+4)}{(x-7)(x+4)} = \frac{1}{3}$$

$$3. \lim_{x \rightarrow -4^+} \frac{g(x)}{h(x)} = \lim_{x \rightarrow -4^+} \frac{x+4}{-2|x+4|} = -\frac{1}{2}$$



The graph of $f(x)$, shown above, has a vertical asymptote at $x = 6$. Use the graph to evaluate the following limits. If a limit does not exist, write DNE.

$$1. \lim_{x \rightarrow 2} f(x) = -2$$

$$2. \lim_{x \rightarrow -1} f(x) \text{ DNE}$$

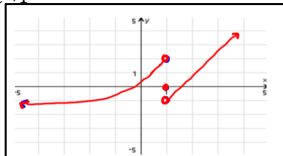
$$3. \lim_{x \rightarrow -1^-} f(x) = -2$$

$$4. \lim_{x \rightarrow 6} f(x) \text{ DNE}$$

Topic: Limits

Sketch a graph that meets the following guidelines.

$$1. \lim_{x \rightarrow 1^-} g(x) = 2, \lim_{x \rightarrow 1^+} g(x) = -1, g(1) = 0$$



$$2. \lim_{x \rightarrow 2^-} h(x) = -\infty, \lim_{x \rightarrow 2^+} h(x) = +\infty$$

$$\lim_{x \rightarrow -\infty} h(x) = 2, \lim_{x \rightarrow \infty} h(x) = -2$$



1. Which limit from the upper right corner does not exist for the same reason as problem 2 on the upper left corner?

Limit in problem 4

2. Sketch a graph of the limit described in the first problem from the top left corner.



3. Using problem 1 from the bottom left corner, explain the difference between finding $g(1)$ and $\lim_{x \rightarrow 1} g(x)$.

Finding $g(1)$ is finding what the "y"-value equals when the x-value is 1. Finding the limit of $g(x)$ as x approaches 1 is finding what value the "y"-value is approaching as x approaches 1.

*Inspired by Tony Record

1.10 - 1.13, 1.16

AP Calculus AB Unit 1 Notes

1.10 Exploring Types of Discontinuities

1.11 Defining Continuity at a Point

1.12 Confirming Continuity over an Interval

1.13 Removing Discontinuities

1.16 Working with the Intermediate Value Theorem (IVT)

Ex 1:

The graph of the function f is shown. Which of the following statements is false?

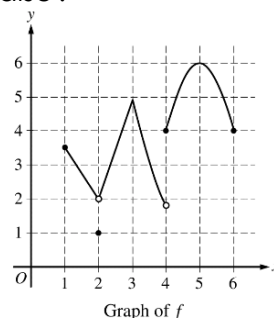
(A) $\lim_{x \rightarrow 2} f(x)$ exists.

(B) $\lim_{x \rightarrow 3} f(x)$ exists.

(C) $\lim_{x \rightarrow 4} f(x)$ exists.

(D) $\lim_{x \rightarrow 5} f(x)$ exists.

(E) The function f is continuous at $x = 3$.



Ex 2:

a) If $f(2) = 4$, can you conclude anything about the $\lim_{x \rightarrow 2} f(x)$? Explain your reasoning.

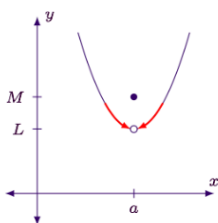
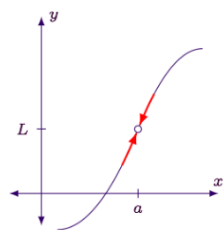
Nothing definitively can be concluded.
 $\lim_{x \rightarrow 2} f(x)$ could be 4.

b) If $\lim_{x \rightarrow 2} f(x) = 4$, can you conclude anything about $f(2)$? Explain your reasoning.

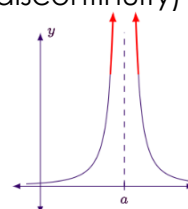
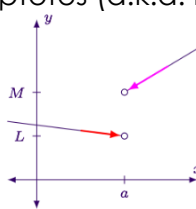
Nothing definitively can be concluded.
 $f(2)$ could be 4.

Types of Discontinuities

removeable – holes



nonremoveable – jumps and vertical asymptotes (a.k.a. infinite discontinuity)



Ex 3: Determine the x -values, if any, where $f(x)$ is discontinuous? For each discontinuity state the x -value, the type of discontinuity, and whether the discontinuity is removable or nonremovable.

a) $f(x) = \frac{x^2 - 1}{x^2 - 4x + 3}$

$x = 1$, hole (removeable)
 $x = 3$, vertical asymptote
(nonremovable)

b) $f(x) = \frac{|x+5|}{x+5}$

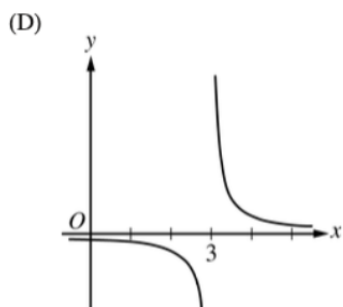
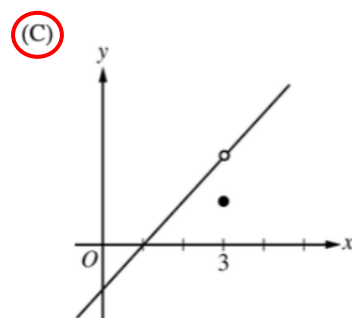
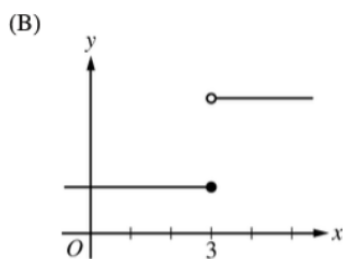
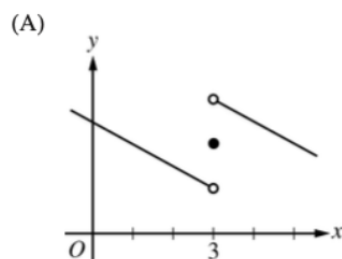
$x = -5$, jump (nonremovable)

1.10 - 1.13, 1.16

AP Calculus AB Unit 1 Notes

Ex 4:

If f is a function that has a removable discontinuity at $x = 3$, which of the following could be the graph of f ?



Ex 5:

Let f be the function given by $f(x) = \frac{(x-2)^2(x+3)}{(x-2)(x+1)}$. For which of the following values of x is f not continuous?

(A) -3 and -1 only

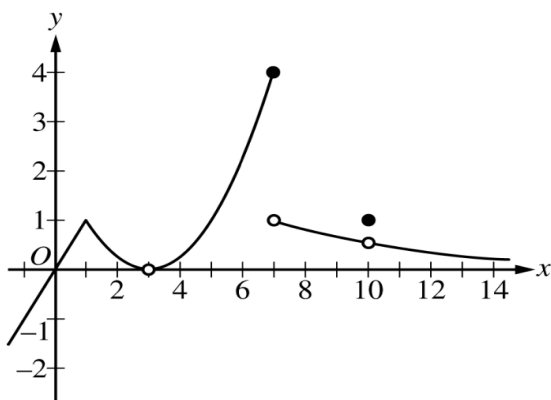
(B) -3 , -1 , and 2

(C) -1 only

(D) -1 and 2 only

(E) 2 only

Ex 6:



Graph of f

The graph of the function f is shown above. At what value of x does f have a jump discontinuity?

(A) 1

(B) 3

(C) 7

(D) 10

1.10 - 1.13, 1.16

Definition of Continuity

Continuity at a Point

A function f is **continuous at c** when these three conditions are met.

1. $f(c)$ is defined.
2. $\lim_{x \rightarrow c} f(x)$ exists.
3. $\lim_{x \rightarrow c} f(x) = f(c)$

Continuity on an Open Interval

A function is **continuous on an open interval (a, b)** when the function is continuous at each point in the interval. A function that is continuous on the entire real number line $(-\infty, \infty)$ is **everywhere continuous**.

Ex 7: Determine if $f(x) = \begin{cases} x + 1, & x \leq 0 \\ x^2 + 1, & x > 0 \end{cases}$ is continuous at $x = 0$. Use the definition of continuity to explain your answer.

1. $f(0) = 0 + 1 = 1$
 2. $\lim_{x \rightarrow 0^-} (x + 1) = 1$ and $\lim_{x \rightarrow 0^+} (x^2 + 1) = 1$, $\lim_{x \rightarrow 0} f(x) = 1$
 3. $f(0) = \lim_{x \rightarrow 0} f(x)$
- $f(x)$ is continuous at $x = 0$.

Ex 8: Determine if $g(x) = \frac{x^3 - 27}{x - 3}$ is continuous at $x = 3$. Use the definition of continuity to explain your answer.

1. $g(3)$ is undefined
- $g(x)$ is discontinuous at $x = 3$.

Ex 9: 2012 AB4 part (c)

The function f is defined by $f(x) = \sqrt{25 - x^2}$ for $-5 \leq x \leq 5$.

(c) Let g be the function defined by $g(x) = \begin{cases} f(x), & -5 \leq x \leq -3 \\ x + 7, & -3 < x \leq 5 \end{cases}$. Is g continuous at $x = -3$? Use the definition of continuity to explain your answer.

1. $g(-3) = f(-3) = 4$
 2. $\lim_{x \rightarrow -3^-} f(x) = 4$ and $\lim_{x \rightarrow -3^+} (x + 7) = 4$, $\lim_{x \rightarrow -3} g(x) = 4$
 3. $g(-3) = \lim_{x \rightarrow -3} g(x)$
- $g(x)$ is continuous at $x = -3$.

Ex 10:

$$f(x) = \begin{cases} x^2 \sin(\pi x) & \text{for } x < 2 \\ x^2 + cx - 18 & \text{for } x \geq 2 \end{cases}$$

Let f be the function defined above, where c is a constant. For what value of c , if any, is f continuous at $x = 2$?

- (A) 2 (B) 7 (C) 9 (D) $4\pi - 4$ (E) There is no such value of c .

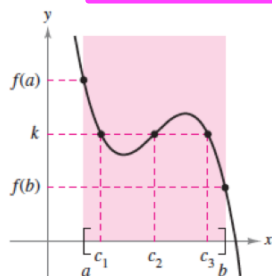
1. $f(2) = 2c - 14$
2. $\lim_{x \rightarrow 2^-} x^2 \sin(\pi x) = 0$
 $\lim_{x \rightarrow 2^+} (x^2 + cx - 18) = 2c - 14$
3. $f(2) = \lim_{x \rightarrow 2} f(x)$
 $2c - 14 = 0$
 $c = 7$

The Intermediate Value Theorem

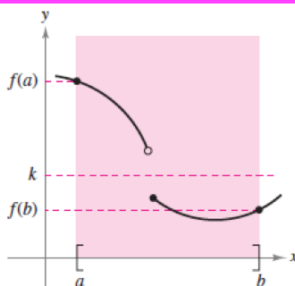
THEOREM 1.13 Intermediate Value Theorem

If f is continuous on the closed interval $[a, b]$, $f(a) \neq f(b)$, and k is any number between $f(a)$ and $f(b)$, then there is at least one number c in $[a, b]$ such that

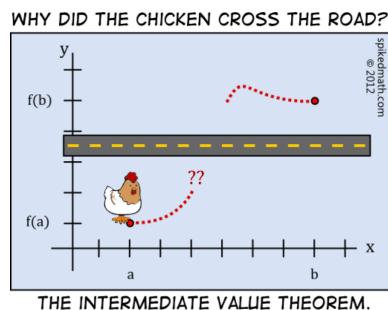
$$f(c) = k.$$



f is continuous on $[a, b]$.
[There exist three c 's such that $f(c) = k$.]



f is not continuous on $[a, b]$.
[There are no c 's such that $f(c) = k$.]



THE INTERMEDIATE VALUE THEOREM.

Ex 11: Use the Intermediate Value Theorem to show $f(x) = x^3 + 2x - 1$ has a zero on the interval $[0, 1]$.

$f(x)$ is a polynomial, $f(x)$ is continuous.

$$f(0) = -1$$

$$f(1) = 2$$

Since $f(0) < 0 < f(1)$, the IVT guarantees $f(x)$ has a zero on $[0, 1]$.

Ex 12:

x	0	4	6	8	13
$f(x)$	3	4.5	3	2.5	4.4

The table above shows selected values of a continuous function f . For $0 \leq x \leq 13$, what is the fewest possible number of times $f(x) = 4$?

- (A) One **(B) Two** (C) Three (D) Four

Ex 13: 2007 AB3 part (a)

$$(a) \quad h(1) = f(g(1)) - 6 = f(2) - 6 = 9 - 6 = 3$$

$$h(3) = f(g(3)) - 6 = f(4) - 6 = -1 - 6 = -7$$

Since $h(3) < -5 < h(1)$ and h is continuous, by the Intermediate Value Theorem, there exists a value r , $1 < r < 3$, such that $h(r) = -5$.

$$2 : \begin{cases} 1 : h(1) \text{ and } h(3) \\ 1 : \text{conclusion, using IVT} \end{cases}$$

The functions f and g are differentiable for all real numbers, and g is strictly increasing. The table above gives values of the functions and their first derivatives at selected values of x . The function h is given by $h(x) = f(g(x)) - 6$.

(a) Explain why there must be a value r for $1 < r < 3$ such that $h(r) = -5$.

1.14 - 1.15

1.14 Connecting Infinite Limits and Vertical Asymptotes

1.15 Connecting Limits at Infinity and Horizontal Asymptotes

Ex 1: State the end behavior. Express your answer using the mathematical notation of a limit.

a) $f(x) = e^x$ $\lim_{x \rightarrow -\infty} f(x) = 0$ $\lim_{x \rightarrow \infty} f(x) = \infty$	b) $g(x) = \frac{2x+3}{x^2-5x+6}$ $\lim_{x \rightarrow -\infty} g(x) = 0$ $\lim_{x \rightarrow \infty} g(x) = 0$	c) $h(x) = \tan^{-1} x$ $\lim_{x \rightarrow -\infty} h(x) = -\frac{\pi}{2}$ $\lim_{x \rightarrow \infty} h(x) = \frac{\pi}{2}$
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Finding Horizontal Asymptotes

To find horizontal asymptotes:

1. consider the graph
2. if given a quotient, consider the growth rate of the numerator and denominator

BOBO

bigger on bottom, zero

BOTN

bigger on top, none (no HA)

EATSDC

exponents (magnitudes) are the same, divide (leading) coefficients

Function Growth Rates (from slowest to fastest)

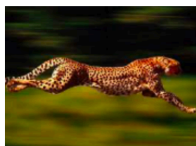


sine/cosine

logarithms

n^{th} roots

polynomials



exponential growth functions

factorials

Ex 2: Find all horizontal asymptotes, if any. Use a graphing utility to confirm your answer.

a) $g(x) = \frac{4x^2-1}{7x-1}$ none	b) $f(x) = \frac{4x-1}{7x^2-1}$ $y = 0$	c) $h(x) = \frac{4x^2-1}{7x^2-1}$ $y = \frac{4}{7}$	d) $y = e^{-x}$ $y = 0$
e) $f(x) = \sin x$ none	f) $y = \frac{\sin x}{x}$ $y = 0$	g) $f(x) = \frac{2^x}{x^2}$ none	h) $h(x) = \frac{\sqrt{9x^2-1}}{2x-1}$ $y = \pm \frac{3}{2}$



1.14 - 1.15

AP Calculus AB Unit 1 Notes

Limits at Infinity

When finding limits at infinity consider the end behavior as the x -values increase and decrease without bound. End behavior is dependent on the existence of a horizontal asymptote.



Ex 3: Evaluate each limit.

a) $\lim_{x \rightarrow \infty} \frac{4x-1}{7x^2-1} = 0$	b) $\lim_{x \rightarrow -\infty} \frac{4x-1}{7x^2-1} = 0$	c) $\lim_{x \rightarrow \infty} e^{-x} = 0$	d) $\lim_{x \rightarrow -\infty} e^{-x} = 0$
e) $\lim_{x \rightarrow \infty} \sin x$ DNE	f) $\lim_{x \rightarrow \infty} \frac{2^x}{x^2} = \infty$	g) $\lim_{x \rightarrow \infty} \frac{\sqrt{9x^2-1}}{2x-1} = \frac{3}{2}$	h) $\lim_{x \rightarrow -\infty} \frac{\sqrt{9x^2-1}}{2x-1} = -\frac{3}{2}$

Ex 4:

$$\lim_{x \rightarrow \infty} \frac{(2x-1)(3-x)}{(x-1)(x+3)} \text{ is}$$

- (A) -3 (B) -2 (C) 2 (D) 3 (E) nonexistent

Ex 5: For which of the following does $\lim_{x \rightarrow \infty} f(x) = 0$?

I. $f(x) = \frac{\ln x}{x^{99}}$	(A) I only
II. $f(x) = \frac{\ln x}{e^x}$	(B) II only
III. $f(x) = \frac{x^{99}}{e^x}$	(C) III only
	(D) I and II only
	(E) I and III only

Ex 6:

What are all horizontal asymptotes of the graph of $y = \frac{5+2^x}{1-2^x}$ in the xy -plane?

- (A) $y = -1$ only
 (B) $y = 0$ only
 (C) $y = 5$ only
 (D) $y = -1$ and $y = 0$
 (E) $y = -1$ and $y = 5$

Horizontal Asymptotes and Limits

The line $y = L$ is a horizontal asymptote of the graph of f when
 $\lim_{x \rightarrow \infty} f(x) = L$ OR $\lim_{x \rightarrow -\infty} f(x) = L$.

1.14 - 1.15

Ex 7: If $\lim_{x \rightarrow \infty} f(x) = 3$, which of the following must be true?

- (A) $\lim_{x \rightarrow -\infty} f(x) = 3$
- (B) $f(x)$ is continuous
- (C) The vertical line $x = 3$ is an asymptote for the graph of $f(x)$.
- (D) The horizontal line $y = 3$ is an asymptote for the graph of $f(x)$.**

Finding Vertical Asymptotes

Let $h(x) = \frac{f(x)}{g(x)}$. If $f(c) \neq 0$ and $g(c) = 0$ then $h(x)$ has a vertical asymptote at $x = c$.

Ex 8: Find all vertical asymptotes, if any.

a) $y = \frac{x+6}{x^2-36}$

$x = 6$

b) $y = \sec x$

$x = \frac{\pi}{2} + \pi n, n \in \mathbb{Z}$

Infinite Limits

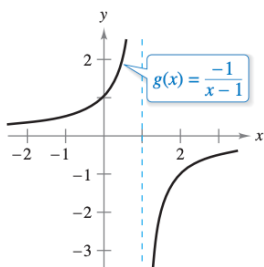
A limit in which $f(x)$ increases or decreases without bound as x approaches c is called an infinite limit.



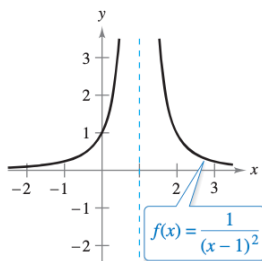
The limit statement $\lim_{x \rightarrow c} f(x) = \infty$ does not mean the limit exists! On the contrary, it tells you how the limit **fails to exist** due to the unbounded nature of $f(x)$ as x approaches c . On the AP exam, you can express your answer as ∞ or DNE.

Ex 9: Evaluate each limit.

a) $\lim_{x \rightarrow 1} f(x)$ **DNE**



b) $\lim_{x \rightarrow 1} f(x) = \infty$



c) $\lim_{x \rightarrow 1} \frac{1}{(x-1)^3}$ **DNE**

d) $\lim_{x \rightarrow 7} \frac{x-9}{x-7}$ **DNE**

e) $\lim_{x \rightarrow 7} \frac{x-9}{(x-7)^2} = -\infty$

f) $\lim_{x \rightarrow 6^+} \frac{x}{x^2-36} = \infty$

g) $\lim_{x \rightarrow 2^-} \frac{x^2+8x+15}{x^2+3x-10} = -\infty$

h) $\lim_{x \rightarrow \frac{\pi}{2}} -2 \sec x = -\infty$

i) $\lim_{x \rightarrow 3^+} \frac{e^x}{x^2-9} = \infty$

Vertical Asymptotes and Limits

The line $x = c$ is a vertical asymptote of the graph of f when

$\lim_{x \rightarrow c^-} f(x) = \pm \infty$ OR $\lim_{x \rightarrow c^+} f(x) = \pm \infty$.

1.14 - 1.15

AP Calculus AB Unit 1 Notes

Ex 10:

The vertical line $x = 2$ is an asymptote for the graph of the function f . Which of the following statements must be false?

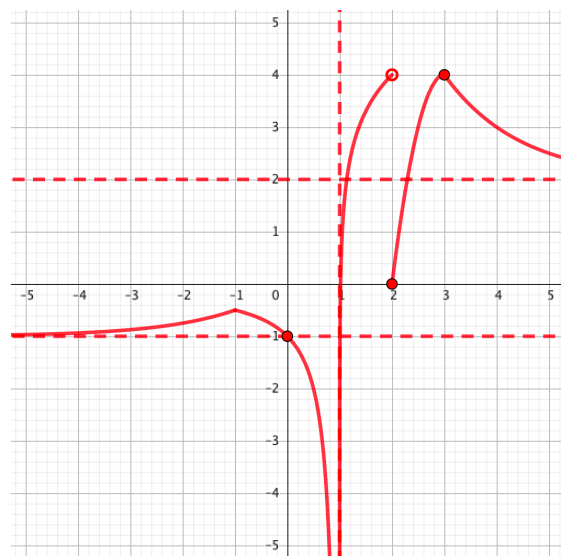
- (A) $\lim_{x \rightarrow 2} f(x) = 0$
- (B) $\lim_{x \rightarrow 2} f(x) = -\infty$
- (C) $\lim_{x \rightarrow 2} f(x) = \infty$
- (D) $\lim_{x \rightarrow \infty} f(x) = 2$
- (E) $\lim_{x \rightarrow \infty} f(x) = \infty$

Ex 11: Evaluate each limit, if possible.

a) <i>mild</i> $\lim_{x \rightarrow \infty} \frac{1}{3 + 5^{1/x}} = \frac{1}{4}$	b) <i>medium</i> $\lim_{x \rightarrow -\infty} \frac{1}{3 + 5^{1/x}} = \frac{1}{4}$	c) <i>spicy</i> $\lim_{x \rightarrow 0} \frac{1}{3 + 5^{1/x}} \text{ DNE}$ $\lim_{x \rightarrow 0^-} \frac{1}{3 + 5^{1/x}} = \frac{1}{3}$ $\lim_{x \rightarrow 0^+} \frac{1}{3 + 5^{1/x}} = 0$
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Ex 12: Graph a possible function that satisfies the conditions below:

- i. $\lim_{x \rightarrow -\infty} f(x) = -1$
- ii. $\lim_{x \rightarrow \infty} f(x) = 2$
- iii. $\lim_{x \rightarrow 1^+} f(x) = -\infty$
- iv. $\lim_{x \rightarrow 2^-} f(x) = 4$
- v. $f(2) = 0$
- vi. $f(0) = -1$
- vii. $f(3) = 4$



**answers will vary*